

Analyyysi I: Limits

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Abstract

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1 Course information

The official course information is available on the University of Helsinki course page.

2 Lecture notes

2.1 01.09.2026

2.1.1 The absolute value function

Definition 2.1. For any real number $x \in \mathbb{R}$, the *absolute value* (or *modulus*) of x , denoted by $|x|$, is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Geometrically, $|x - y|$ is the distance between the points x and y on the real line.

Example 2.2. Fix $y \in \mathbb{R}$ and let $h > 0$. The inequality $|x - y| < h$ can be written in several equivalent ways:

$$\begin{aligned} |x - y| < h &\iff -h < x - y < h \\ &\iff y - h < x < y + h \\ &\iff x \in (y - h, y + h). \end{aligned}$$

Thus, $|x - y| < h$ means that x lies in the open interval of radius h centred at y .

Example 2.3. As a concrete instance of the preceding equivalences,

$$\begin{aligned} |2x - 1| < 3 &\iff -3 < 2x - 1 < 3 \\ &\iff -2 < 2x < 4 \\ &\iff -1 < x < 2 \\ &\iff x \in (-1, 2). \end{aligned}$$

Theorem 2.4 (Triangle and reverse triangle inequalities). *For all $x, y \in \mathbb{R}$,*

$$\begin{aligned} |x + y| &\leq |x| + |y|, \\ ||x| - |y|| &\leq |x - y|. \end{aligned}$$

Proof. By definition 2.1,

$$\begin{aligned} -|x| &\leq x \leq |x|, \\ -|y| &\leq y \leq |y|. \end{aligned}$$

Adding the inequalities gives

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

which is equivalent to $|x + y| \leq |x| + |y|$.

Applying this triangle inequality to $x = (x - y) + y$ gives

$$|x| - |y| \leq |x - y|.$$

Interchanging x and y similarly gives $|y| - |x| \leq |x - y|$. Therefore

$$-|x - y| \leq |x| - |y| \leq |x - y|,$$

which is equivalent to the reverse triangle inequality. □

2.2 03.09.2026

2.2.1 More on absolute values

Theorem 2.5. *We have*

$$|xy| = |x||y|$$

for all $x, y \in \mathbb{R}$.

Proof. We prove by splitting into four cases. Firstly, if $x, y \geq 0$, we have $xy \geq 0$, so $|xy| = xy$. But in this case also both $|x| = x$ and $|y| = y$, so $|xy| = xy = |x||y|$. Secondly, if $x, y < 0$, we have $xy > 0$, $|xy| = xy = (-x)(-y) = |x||y|$. Now suppose $x < 0$ and $y \geq 0$. Then $xy \leq 0$, and hence $|xy| = -xy = (-x)y = |x||y|$. The case $x \geq 0, y < 0$ follows from commutativity of multiplication of real numbers. $\square$

Example 2.6. Let $x \in (1, 3)$ and $y \in (4, 8)$. Find an upper bound for

$$|x + y - 8|.$$

Kolmioepäyhtälön nojalla saadaan

$$\begin{aligned} |x + y - 8| &= |(x - 2) + (y - 6)| \\ &\leq |x - 2| + |y - 6| \\ &< 1 + 2 \\ &= 3. \end{aligned}$$

Thus we have found that 3 is an upper bound for $|x + y - 8|$.

Example 2.7. Let $x \in (2, 6)$ and $y \in (2, 8)$. Find an upper bound for

$$|xy - 20|.$$

We have $|x - 4| < 2$ and $|y - 5| < 3$. We have

$$\begin{aligned} |xy - 20| &= |xy - 4y + 4y - 20| \\ &\leq |xy - 4y| + |4y - 20| \\ &= |y||x - 4| + 4|y - 5| \\ &\leq 8 \cdot 2 + 4 \cdot 3 \\ &= 28. \end{aligned}$$

It follows that 28 is an upper bound for $|xy - 20|$.

Example 2.8. Find a positive real number m such that

$$|x^2 - 16| \leq m|x - 4|,$$

where $x \in (0, 10)$.

When $x = 4$, we get $0 \leq 0$, which holds with every $m \in \mathbb{R}$. Now suppose $x \neq 4$. We simplify the inequality and obtain

$$|x + 4| \leq m.$$

Since $x \in (0, 10)$, $|x + 4| \in (4, 14)$. We have thus obtained

$$|x + 4| \leq 14,$$

so we can choose $m = 14$.

2.2.2 Field and ordering properties of $\mathbb{R}$

Definition 2.9. $(F, +, \cdot)$ is a field if and only if

1. The algebraic structure $(F, +)$ is an *abelian group*.
2. The algebraic structure $(F^*, \cdot)$ is an *abelian group*, where $F^* = F \setminus \{0_F\}$.
3. The operation $\cdot$ distributes over $+$.

Theorem 2.10. $\mathbb{R}$ is a field.

Proof. Consider the usual addition and multiplication on $\mathbb{R}$.

First, $(\mathbb{R}, +)$ is an abelian group. Addition is closed, associative, and commutative; 0 is an additive identity; and every $x \in \mathbb{R}$ has the additive inverse $-x$.

Next, $(\mathbb{R} \setminus \{0\}, \cdot)$ is an abelian group. Multiplication is associative and commutative, and 1 is its identity element. If $x \neq 0$, then $x^{-1} = 1/x$ is a nonzero real number satisfying

$$x \cdot x^{-1} = 1.$$

The product of two nonzero real numbers is nonzero, so multiplication is closed on $\mathbb{R} \setminus \{0\}$.

Finally, multiplication distributes over addition:

$$x(y + z) = xy + xz$$

for all $x, y, z \in \mathbb{R}$. Therefore $(\mathbb{R}, +, \cdot)$ is a field. $\square$

Definition 2.11 (Ordered field). An *ordered field* is a field $(F, +, \cdot)$ equipped with a relation $<$ such that, for all $x, y, z \in F$,

1. exactly one of $x < y$, $x = y$, or $x > y$ holds;
2. if $x < y$ and $y < z$, then $x < z$;
3. if $x < y$, then $x + z < y + z$; and
4. if $0_F < x$ and $0_F < y$, then $0_F < xy$.

Here $x > y$ means $y < x$. The first two conditions state that $<$ is a strict total order, while the last two state that the order is compatible with the field operations.

Theorem 2.12. The field $(\mathbb{R}, +, \cdot)$, equipped with the usual order $<$, is an ordered field.

Proof. We have already shown that $(\mathbb{R}, +, \cdot)$ is a field. The usual strict order on $\mathbb{R}$ satisfies trichotomy and transitivity, so it satisfies the first two ordering axioms.

Suppose that $x < y$. Then $0 < y - x$, and for every $z \in \mathbb{R}$,

$$(y + z) - (x + z) = y - x > 0.$$

Hence $x + z < y + z$. Furthermore, if $0 < x$ and $0 < y$, then the usual sign rule for multiplication gives $0 < xy$. Thus the order is compatible with addition and multiplication, and $\mathbb{R}$ is an ordered field. $\square$

Theorem 2.13 (Uniqueness of additive identity). *The identity element 0_F of the abelian group $(F, +)$ is unique.*

Proof. Suppose that 0_F and $0'_F$ are both additive identity elements of $(F, +)$. Since 0_F is an additive identity,

$$0_F + 0'_F = 0'_F.$$

Since $0'_F$ is also an additive identity,

$$0_F + 0'_F = 0_F.$$

Therefore $0_F = 0'_F$. □

Theorem 2.14 (Cancellation law). *Let $x, y, z \in \mathbb{R}$. If $x + y = x + z$, then $y = z$.*

Proof. Using the additive identity, additive inverse, associativity, and the assumption $x + y = x + z$, we obtain

$$\begin{aligned} y &= 0 + y \\ &= ((-x) + x) + y \\ &= (-x) + (x + y) \\ &= (-x) + (x + z) \\ &= ((-x) + x) + z \\ &= 0 + z \\ &= z. \end{aligned}$$

□

2.3 08.09.2026

2.3.1 Sequences

Definition 2.15. Let X and Y be sets. A function $f: X \rightarrow Y$ is a relation $f \subseteq X \times Y$ such that, for every $x \in X$, there exists exactly one $y \in Y$ with $(x, y) \in f$. This unique y is denoted by $f(x)$.

The set X is the *domain* of f , and Y is its *codomain*. The set

$$f(X) = \{f(x) : x \in X\}$$

is its *range* or *image*.

Definition 2.16. A sequence of real numbers is a function $x: \mathbb{N} \rightarrow \mathbb{R}$. For $n \in \mathbb{N}$, the value $x(n)$, usually denoted by x_n , is the n -th term of the sequence. We denote the sequence by $(x_n)_{n \in \mathbb{N}}$, or simply by (x_n) .

2.3.2 Limit of a sequence

Definition 2.17. Let (x_n) be a sequence of real numbers. We say that (x_n) *converges* to $a \in \mathbb{R}$ if, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $n \in \mathbb{N}$ with $n > N$,

$$|x_n - a| < \epsilon.$$

In this case, we write $x_n \rightarrow a$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} x_n = a$.

Let us look at some examples.

Example 2.18. Let (x_n) be defined by

$$x_n = \frac{2n}{n+1}.$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 2.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{2}{\epsilon}$. Now for every $n > N$, we have

$$\begin{aligned} |x_n - 2| &= \left| \frac{2n}{n+1} - 2 \right| \\ &= \frac{2}{n+1} < \frac{2}{n} < \frac{2}{\frac{2}{\epsilon}} = \epsilon. \end{aligned}$$

Thus $x_n \rightarrow 2$ when $n \rightarrow \infty$.

Example 2.19. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_7 = 42, \\ x_n &= 7 + \frac{1}{n}, \quad n \geq 8. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 7.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max \left\{ 8, \frac{1}{\epsilon} \right\}$. Now for every $n > N$, we have

$$|x_n - 7| = \frac{1}{n} < \epsilon.$$

Thus $x_n \rightarrow 7$ as $n \rightarrow \infty$.

Let us now tackle a simple divergence proof.

Example 2.20. Let us show that the sequence x_n defined by $x_n = n$ diverges. Assume that the sequence converged to some $a \in \mathbb{R}$. Taking $\epsilon = 1$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - a| < \epsilon = 1.$$

Choose $n \in \mathbb{N}$ such that

$$n > \max \{N, |a| + 1\}.$$

By the triangle inequality,

$$n = |n| = |(n - a) + a| \leq |n - a| + |a|.$$

Subtracting $|a|$ from both sides and using $n > |a| + 1$, we obtain

$$|x_n - a| = |n - a| \geq n - |a| > 1,$$

contradicting the assumption $|x_n - a| < 1$. Therefore (x_n) diverges.

Example 2.21. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_{42} = 1, \\ x_n &= \frac{n}{n - 42}, \quad n \geq 43. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 1.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$N > \max \left\{ 84, \frac{84}{\epsilon} \right\}.$$

Now, for every $n > N$, we have $n > 84$, and hence $\frac{n}{2} - 42 > 0$. Therefore

$$\begin{aligned} |x_n - 1| &= \left| \frac{n}{n - 42} - 1 \right| \\ &= \frac{42}{n - 42} \\ &= \frac{42}{\frac{n}{2} + \frac{n}{2} - 42} \\ &< \frac{84}{n} < \epsilon. \end{aligned}$$

Thus $x_n \rightarrow 1$ as $n \rightarrow \infty$.